ACADEMIC CENTRAL

PRACTICE EXAMINATION

# NC(V) LEVEL 2

# MECHATRONICS

## PAPER 1 · PRACTICE E

**MARKS: 100** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 2. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Energy and technical systems, bimetal thermostats and gears, sensors, switches and logic, pneumatics and the service unit, electrical protection and a start/stop control circuit, and fault-finding and maintenance.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw all symbols and circuits neatly, using a ruler.
5. Use only a black or blue pen. Write neatly and legibly.

## QUESTION 1: ENERGY AND TECHNICAL SYSTEMS

**1.1** Name FIVE devices used in mechatronic systems that change electrical energy into another form of energy, and state the form of energy each one produces. (5)

**1.2** Explain the input, process and output of a technical system, using a drinks vending machine as an example. (6)

**1.3** Explain how a bimetal thermostat switches off a heater when the temperature becomes too high. (4)

**1.4** Explain the function of a gearbox, and name ONE type of gear that changes the direction of rotation through 90°. (5)

**[20]**

## QUESTION 2: SENSORS, SWITCHES AND LOGIC

**2.1** Define the following terms:

2.1.1 Sensor (2)

2.1.2 Actuator (2)

2.1.3 Logic gate (2)

**2.2** Draw a circuit in which two switches (S1 and S2) and a lamp (E1) form an AND function, and explain how it works. (4)

**2.3** Draw the symbol of EACH of the following:

2.3.1 Normally open (NO) push button (2)

2.3.2 Normally closed (NC) push button (2)

2.3.3 Roller-operated limit switch (2)

**2.4** Write the following acronyms in full:

2.4.1 PLC (1)

2.4.2 HMI (1)

2.4.3 PPE (1)

2.4.4 DC (1)

**[20]**

## QUESTION 3: PNEUMATICS

**3.1** Give FOUR advantages and TWO disadvantages of using compressed air (pneumatics) in a mechatronic system. (6)

**3.2** Name the THREE parts of a pneumatic service unit and give the function of each part. (6)

**3.3** Draw the symbol of a single-acting cylinder with a spring return. (3)

**3.4** What is a shuttle valve (OR valve) mainly used for? (3)

**3.5** A cylinder has a piston area of 0,002 m² and is supplied with air at 600 kPa. Calculate the force it can produce. (2)

**[20]**

## QUESTION 4: ELECTRICAL CONTROL AND SAFETY

**4.1** Explain the difference between a fuse and a circuit breaker. (4)

**4.2** What is the function of an earth leakage unit? (2)

**4.3** Draw a start/stop control circuit for a motor contactor K1. The circuit must have an emergency stop, a STOP button, a START button, a holding (self-latching) contact and an overload contact. (8)

**4.4** Why is 24 V DC used for control circuits in mechatronic systems? (2)

**4.5** List FOUR safety rules to follow before working on electrical equipment. (4)

**[20]**

## QUESTION 5: FAULT-FINDING AND MAINTENANCE

**5.1** List FOUR faults that can occur in pneumatic cylinders. (4)

**5.2** Explain the half-split (half-cut) method of fault-finding. (4)

**5.3** Draw the flowchart symbol for EACH of the following: start/end; process; decision; input/output. (4)

**5.4** Explain the difference between preventive maintenance and corrective maintenance. (4)

**5.5** List FOUR items of information that should be recorded on a maintenance record card. (4)

**[20]**

**TOTAL: 100**

**END OF QUESTION PAPER**